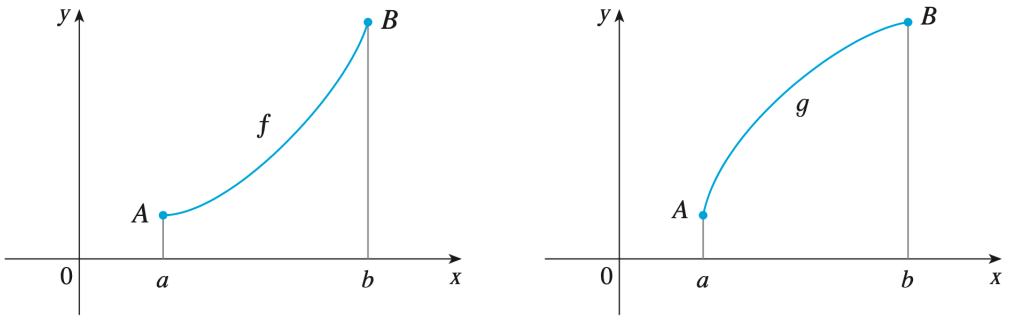
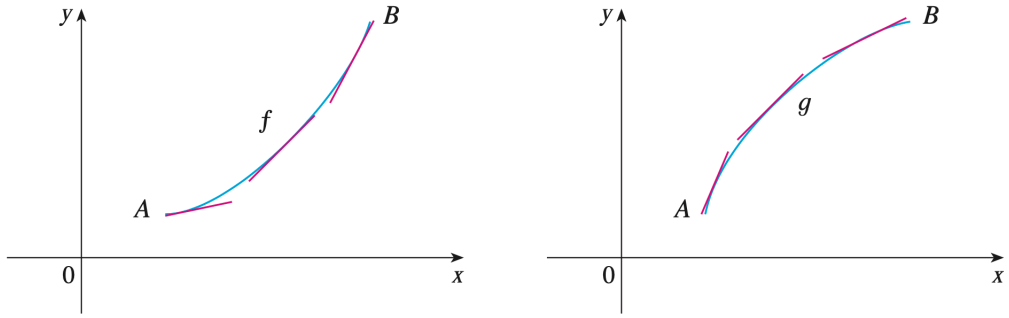
Corollary 3 of the Mean Value Theorem says that if $f'(x) > 0$ on an interval then f is increasing on that interval. Similarly, if $f'(x) < 0$ on an interval then f is decreasing on that interval. Consider now the two graphs shown below:



Both graphs join point A to point B, and both are increasing on the interval $[a,b]$, but they look different because they bend in different directions. How can we distinguish these two types of behavior?

1. In the figure below we sketch the tangent line at a few points on each graph. What do you notice about the behavior of the tangent lines in each case?



Definition. Let f be differentiable on an interval I.

(a) We say that f is *concave up* on I if at each point of I the tangent line to the graph lies below the graph.

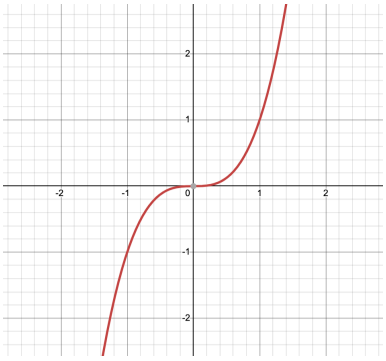
(b) We say that f is *concave down* on I if at each point of I the tangent line to the graph lies above the graph.

Theorem. (Test for Concavity) Let f be twice differentiable on an interval I.

(a) If $f''(x) > 0$ for all $x \in I$, then f is concave up on I.

(b) If $f''(x) < 0$ for all $x \in I$, then f is concave down on I.

Example. For the function $f(x) = x^3$, $f''(x) = (3x^2)' = 6x$, so $f''(x) < 0$ when $x < 0$ and $f''(x) > 0$ when $x > 0$. The graph below confirms that f is concave down on the interval $(-\infty, 0)$ and concave up on $(0, \infty)$.



In the example above, the concavity changed at the point $x = 0$.

Definition. A point $(a, f(a))$ on the graph of a function $y = f(x)$ is called an *inflection point* if

(a) f is continuous at a, and

(b) the concavity changes at a.

It follows from the Test for Concavity that a continuous function can only change concavity at a point x where $f''(x) = 0$ or $f''(x)$ is undefined. These are precisely the critical points of the function f'. To avoid confusion, we will call these "potential inflection points", rather than "critical points".

Definition. A *potential inflection point* of a continuous function f is a point x at which $f''(x) = 0$ or $f''(x)$ is undefined.

If f is a continuous function on an interval I, and $c \in I$ a potential inflection point of f. Then it follows immediately from the Test for Concavity that f changes concavity at c if and only if f'' changes sign at c.

5. For the function $f(x) = x^3 - 6x^2 + 9x + 30$, determine all intervals where f is concave up and all intervals where f is concave down. List all inflection points for f.

6. For $f(x) = -x^3 + \frac{3}{2}x^2 + 18x$, find all intervals where f is concave up and all intervals where f is concave down. List any inflection points.

Second Derivative Test

We note that the second derivative can also be used to find local extrema in some cases. This is called the *Second Derivative Test* and can sometimes be a simpler method than the First Derivative Test.

Definition. A point where $f'(x) = 0$ is called a *stationary point* of f.

Note that every stationary point is a critical point, but not conversely (since a point where $f'(x)$ does not exist is a critical point, but not a stationary point).

Theorem. (Second Derivative Test) Suppose f'' is continuous on an interval containing a stationary point c.

(a) If $f''(c) > 0$, then f has a local minimum at c.

(b) If $f''(c) < 0$, then f has a local maximum at c.

7. Find all local extrema of the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$.

8. Use the Second Derivative Test to find the locations of all local extrema for $f(x) = x^5 - 5x^3$.